

CAMPBREAK · CALCULATION NOTE

CampBreak sizing calculations

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DOCUMENT CONTROL

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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-03	Amish Chadha	Amish's R9 decision (CBK-DDR-003): check foot valve, suction hose left coupled, go-when-two-arrive rule; R9 timeline recomputed with the delivery hose fill time added; cart mass and cost updated
0.1	2026-10-03	Amish Chadha	First calculation note at TRL 3, on the constructable design (CBK-DDR-002)

CampBreak sizing calculations

On paper the kit meets nine of its eleven requirements. The hand pump gives about 20.4 L/min with two people pumping at a sustainable 47 W each, the 6 mm jet reaches about 6.9 m, the alarms are loud enough and fast enough, and two people pull the loaded cart up a 10 % slope with about 92 N each. Two requirements are not shown: R2 (no false alarm while cooking) cannot be calculated and needs a cooking trial at TRL 4, and R9 (water on target within 3 min for a shelter 100 m away) is still **not met on paper**, at 3.2 min (191 s, 11 s over). Amish's R9 decision of 2026-10-03 (CBK-DDR-003: check foot valve, suction hose left coupled to the pump, "go when two arrive" with a 30 s gather) cuts the time by about 62 s on a like-for-like basis. This revision also adds the time to fill the empty 30 m delivery hose (26 s), which version 0.1 left out. A shelter at most 70 m from a station, the siting margin kept from CBK-DDR-001 (D9), gets water in 2.7 min. The kit costs about USD 1,374 against a value-engineering target of USD 1,600.

Every figure comes from `docs/04-calcs/sizing.py`, which reads its geometry from `cad/src/model.py` and its prices from `bom/bom.csv`, and writes `docs/04-calcs/results.csv`. Inputs marked (est.) are estimates to be confirmed by measurement at TRL 4.

Safety: The pump and hose are a pressurised water system. A shut nozzle with two people pumping could reach several bar; the 4 bar relief valve (CBK-DDR-001, D7) caps it, and the hose is rated at 10 bar. The station battery is sealed lead-acid: it can deliver high short-circuit current and vents hydrogen if overcharged. The kit is for small, early fires only; people leave first, and water is never used on burning oil or live wiring.

■ A. Pump, hose and nozzle (R7, R8)

Assumptions

Table 1. Inputs for the water line.

Input	Value	Note
Pump displacement	0.40 L per double stroke	(est.) to confirm on the pump bought
Pumping rate	60 double strokes per minute	(est.) steady pace for two people
Volumetric efficiency	0.85	(est.) cast semi-rotary pump
Mechanical efficiency	0.50	(est.) lever, vanes and seals
Jet	6 mm, discharge coefficient 0.96	Nozzle orifice to confirm when bought
Delivery hose	19 mm bore, 30 m, roughness 0.01 mm	Plus 1.2 m of 19 mm connecting hose
Delivery fittings	Loss factor 4.0	(est.) swivel, tee and couplings
Suction	25 mm, 4 m, loss factor 7.0, lift 1.5 m	(est.) spring check foot valve, strainer, elbow and coupling (CBK-DDR-003); worst case from a low open container
Nozzle height	1 m above the pump	Slope or raised arm
Reach factor	0.50 of the drag-free range at 30°	(est.) to measure
Sustained effort	75 W per person for 10 min	(est.)

Method

Flow is displacement x rate x volumetric efficiency. Jet speed comes from the flow and the orifice; nozzle pressure is half the density times the jet speed squared. Hose friction uses the Darcy factor from the Swamee-Jain equation. The pump head is the sum of nozzle pressure, friction, fittings, lift to the nozzle and suction losses. Shaft power is hydraulic power over mechanical efficiency; the lever force follows from the work per double stroke over four 40° swings at a 610 mm grip radius.

Results

Table 2. Water line at two people pumping.

Quantity	Result
Flow at the nozzle	20.4 L/min (R7 target 20)
Jet speed	12.5 m/s
Pressure at the nozzle	78 kPa
Hose friction, 30 m	29.9 kPa at 1.2 m/s, Reynolds number about 22,800
Pump outlet pressure	122 kPa (1.2 bar), well under the 4 bar relief setting
Suction side	17 kPa below atmosphere; 8.3 m of net positive suction head available
Hydraulic power	47 W; shaft power 95 W, or 47 W per person
Force at the grip	56 N in total, about 28 N per person
Force to reach the relief setting against a shut nozzle	94 N in total, so two people can reach it
Jet reach	6.9 m (13.9 m drag-free); R8 target 6 m
One 200 L drum	about 9.8 min of pumping

Effort is well within the 75 W per person that can be kept up for 10 minutes, so R7 is met with margin on effort, though only with 2 % margin on flow. The flow depends directly on the pump's displacement, which must be confirmed on the pump bought (CBK-DEC-001, to confirm).

Figure 1 ([media/flow.png](#)) shows the pressure falling from the pump to the nozzle; all values are estimates.

■ B. Heat alarm (R1 to R5)

Table 3. Alarm inputs (est. unless stated).

Input	Value
Threshold	8 °C per minute over 60 s, plus 57 °C fixed (decided, CBK-DDR-001 D2)
Sample interval	10 s; 1 mA for 1 ms per sample
Sleep current	3 µA
Radio listen for station broadcast	every 4 s, 10 mA for 1.5 ms
Daily check-in	one 60 ms transmission at 40 mA
Cells	3 x AA alkaline, 2.4 Ah derated to 70 %, with a 30 min sounding reserve
Sounder	95 dB(A) at 1 m
Alert	100 ms transmission with three retries 1 s apart

Table 4. Alarm results.

Requirement	Result	Status
R1 rate of rise	8 °C per minute set in firmware; worst-case extra delay one 10 s sample	Met by design
R2 cooking	Not calculable on paper	To show at TRL 4
R3 sound at 3 m	$95 - 20 \log_{10}(3) = 85.5 \text{ dB(A)}$	Met (est.), 0.5 dB margin
R4 relay time	3.1 s alarm to station (with retries) plus 4.1 s for the station broadcast to reach every alarm; 7.2 s worst case	Met (est.)
R5 battery life	Average 6.9 μA ; 331 months on paper, so cell shelf life (about 5 years) is the limit	Met (est.)

R3 has little margin: a sounder of at least 95 dB(A) at 1 m is specified in the BOM, and a louder one is the first fix if the TRL 4 measurement falls short.

■ C. Cart mass and balance (R6)

Steel parts are weighed from the model volumes at 7,850 kg/m³; bought parts use catalogue estimates (pump 14 kg, wheels 5.5 kg each, reel drum 9 kg, wound hose 11.4 kg, brass check foot valve 1.0 kg). Since CBK-DDR-003 the cart is stowed primed: the pump body (about 0.6 L, est.) and the 4 m suction hose (2.0 L) hold 2.6 kg of water, counted at the middle of the suction hose run.

Table 5. Cart mass and balance.

Quantity	Result
Loaded cart mass, stowed primed	94.6 kg (est.), of which 2.6 kg is water in the pump and suction hose
Centre of mass	6 mm left of the centre line, 107 mm in front of the axle and 397 mm up
Load on the front feet when parked	126 N
Lift at the pull bar to raise the feet	73 N
Tipping sideways	at 43°
Overall size	850 wide x 1,744 long x 1,067 tall

■ D. Pulling effort and speed (R6)

Rolling resistance 0.10 on unpaved ground (est.), slope 10 %, two people. Pull force is the weight times the sine of the slope plus rolling resistance times the cosine: 185 N in all, 92 N each, below a sustainable 100 N per person (est.). At 1.0 m/s, 100 m takes 1.7 min. The cart is 850 mm wide against a 900 mm limit. R6 is met (est.).

■ E. Time to water on target (R9)

Amish decided on 2026-10-03 (CBK-DDR-003, decision 1A): "1A 2A 3A 4A 5A 6A 7A 8A 9A 10A 11A". For CampBreak, 1A means three changes, with the 3 min target and the 70 m siting margin kept:

1. A brass spring-loaded foot valve with a positive-seal check holds the 4 m suction hose and the pump full of water between uses, so the pump does not need priming.

2. The suction hose stays coupled to the pump inlet by its cam-lever coupling and lies along the left rail, so nothing has to be coupled at the fire.
3. The drilled rule "go when two arrive": the first two volunteers at the station take the cart at once, at most 30 s after the siren. Others follow to the fire.

On arrival the two split the work. One lifts the suction hose off the cart, drops the foot valve in the water and starts pumping. The other runs out the hose from the reel and aims. Once pumping starts, the empty delivery line (30 m of 19 mm hose and 1.2 m of connecting hose, 8.85 L) must fill before water leaves the nozzle: 26 s at 20.4 L/min. Version 0.1 of this note left the fill time out.

Table 6. Time from alarm to water on target.

Stage	100 m (R9)	70 m (siting margin)	Note
Alarm to siren	7 s	7 s	Section B
Gather: go when two arrive	30 s	30 s	Drilled; at most 30 s
Pull the cart out	15 s	15 s	(est.)
Walk at 1.0 m/s	100 s	70 s	Section D
Drop the coupled suction hose in the water	10 s	10 s	(est.) no coupling, no priming
First strokes to lift water	3 s	3 s	(est.) pump and suction already full
Fill the delivery line	26 s	26 s	8.85 L at 20.4 L/min
Run out the hose and aim	25 s, at the same time	25 s, at the same time	(est.) second person; not on the critical path
Total	191 s (3.2 min)	161 s (2.7 min)	

R9 is **still not met on paper** at 100 m: 191 s against 180 s, 11 s over. At 70 m, the siting margin kept from CBK-DDR-001 (D9), it is met with 19 s to spare.

Table 7. What the decision changed, at 100 m.

Case	Time
Version 0.1 timeline (60 s muster, 30 s set-up, 15 s priming), as published	227 s (3.8 min)
Version 0.1 timeline with the delivery line fill added	253 s (4.2 min)
Decided design (CBK-DDR-003)	191 s (3.2 min)
Decided design with the delivery hose also stowed full (not decided; see CBK-DEC-001)	177 s (2.95 min)

On a like-for-like basis the decision saves 62 s. The remaining 11 s is the delivery line fill. If the 30 m hose were also stowed full of water behind the shut nozzle, water would leave the nozzle within the first strokes and the 100 m time would be 177 s, met with 3 s to spare. That would add 8.9 kg to the cart and needs Amish's decision; it is posed in the design decisions register.

Until the next volunteers arrive, one person pumps alone. Holding 60 double strokes a minute needs about 95 W at the shaft and 56 N at the grip from that person, more than the 75 W assumed sustainable for 10 minutes (est.), but for the minute or two until the others arrive it is a reasonable effort. The drill (build plan, section 5) checks it.

■ F. Station roof and posts

Wind 30 m/s (est., a strong monsoon gust), net uplift coefficient 1.2 on the 3.1 m² roof: about 1,990 N of uplift, 500 N per post. Each concrete collar (300 mm across, 600 mm deep) weighs about 975 N, so uplift is resisted with a factor of about 2. A horizontal load of 210 N gives a bending stress of about 10 MPa at the foot of each 60 x 60 x 3 mm post, far below the 235 MPa yield of S235. The four-post frame weighs about 86 kg of steel plus 14 kg of roof sheet.

■ G. Hose reel capacity

30 m of hose with a 28 mm outside diameter needs 18.5 L of space; the drum (200 mm core, 440 mm wound diameter, 210 mm wide, packing 0.785) offers 19.9 L, a ratio of 1.08. The hose fits with little room to spare, so it must be wound neatly (build plan, step 10).

■ H. Cost (R11)

Table 8. Cost of one block kit, from bom/bom.csv.

Part of the kit	USD
24 heat alarms	420
Station (alarm box, frame and roof, sign, drum)	317
Hose cart	637
Total	1,374

Value-engineering target: USD 1,600. Estimated cost of the constructable design: USD 1,374 (USD 226 under the target). CBK-DDR-003 added USD 19: the brass check foot valve (USD 12 more than the plain foot valve), two rubber hose straps (USD 5) and the laminated drill card (USD 2).

■ Results against every requirement

Table 9. Results (from results.csv).

ID	Requirement	Target	Result	Status
R1	Rate-of-rise threshold	8 °C/min	8 °C/min over 60 s, plus 57 °C fixed	Met by design
R2	No alarm in 1 h of cooking at 1 m	No alarm	Not calculable on paper	To show at TRL 4
R3	Sound at 3 m	At least 85 dB(A)	85.5 dB(A)	Met (est.)
R4	Relay to station and all alarms	10 s or less	7.2 s worst case	Met (est.)
R5	Battery life	12 months or more	Cell shelf life, about 5 years	Met (est.)
R6	Cart on a 10 % slope, 100 m	3 min or less, width 0.9 m or less	1.7 min; 92 N per person; 850 mm wide	Met (est.)
R7	Flow for 10 min	20 L/min or more	20.4 L/min; 47 W per person	Met (est.)
R8	Jet reach	6 m or more	6.9 m	Met (est.)
R9	Water on target 100 m away	3 min or less	3.2 min (191 s) at 100 m; 2.7 min at 70 m (siting margin)	Not met on paper at 100 m, 11 s over; met at 70 m
R10	Repairable locally	Hand tools, market parts	Every wear part is a market item	Met by design
R11	Cost	Value-engineering target USD 1,600	USD 1,374	USD 226 under the target